

GAIRA V7 — Phase 03

Emergent Biochemical Theme Discovery

Figure report — all 13 figures with captions

Status	COMPLETE — themes derived FROM the frozen CSMs
Frozen atlas	09ed804a40836f4a05a91ba10900cded
Frozen LSM registry	208482d6f7178b5b8f16cace91be55b0
Frozen CSM dictionary	0b4aa550ccefed3edabdbde5bae11c8d
Membership model	archetypal
K (= BSV dimension)	5
Themes accepted / rejected	4 / 1
Bridge CSMs	15 of 49 — left as bridges
Poorly-explained CSMs	9 of 49 — left unplaced
Bootstrap stability	0.727
Leave-one-out stability	0.738
Ontology agreement (post hoc)	AMI 0.157, p = 0.0000
Theme layer adds value over CSMs	True
Theme fingerprint	f54d4835ffdf8aa2d50a4a203da0e8f4
Completed	2026-08-06 15:39:31 UTC

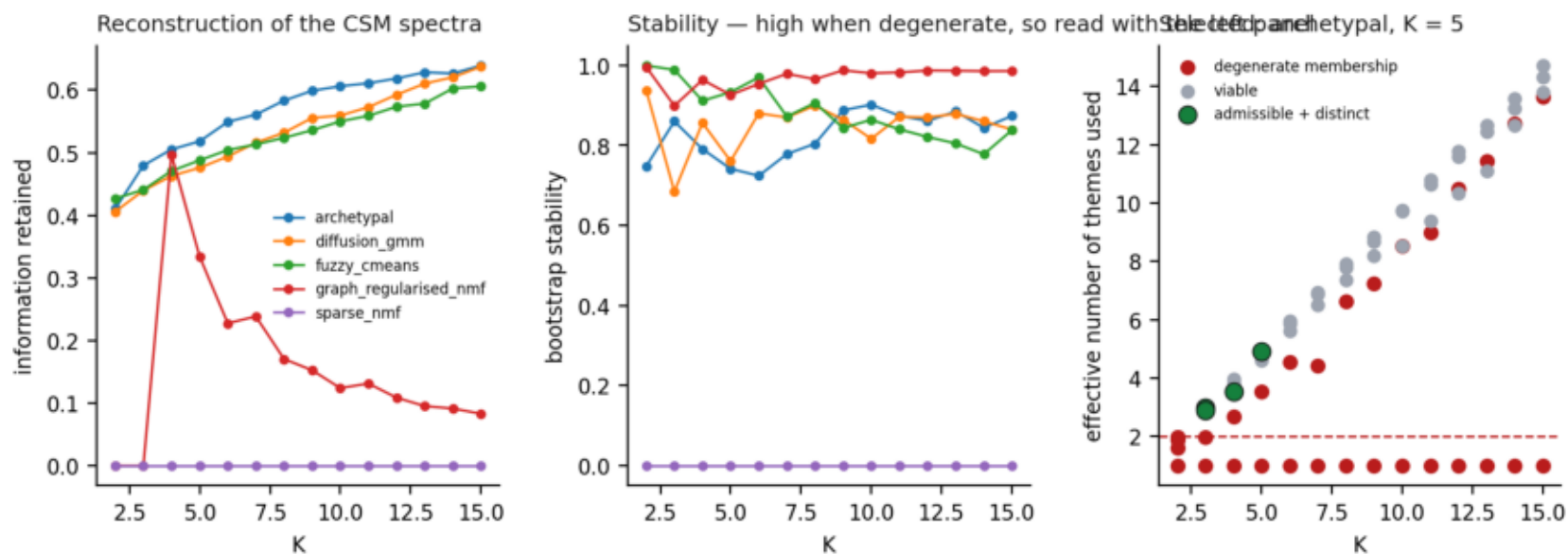
Committed sources of record: reports/PHASE_03_REPORT.md and reports/PHASE_03_SCIENTIFIC_AUDIT.md
No chemistry label was visible during discovery; labels were revealed once, after K was selected and validated.
Figures are embedded from the committed PNGs, so this PDF cannot drift from the run that produced them.

Figures

- 01 model and k selection
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- 09 theme gradients
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Figure 01

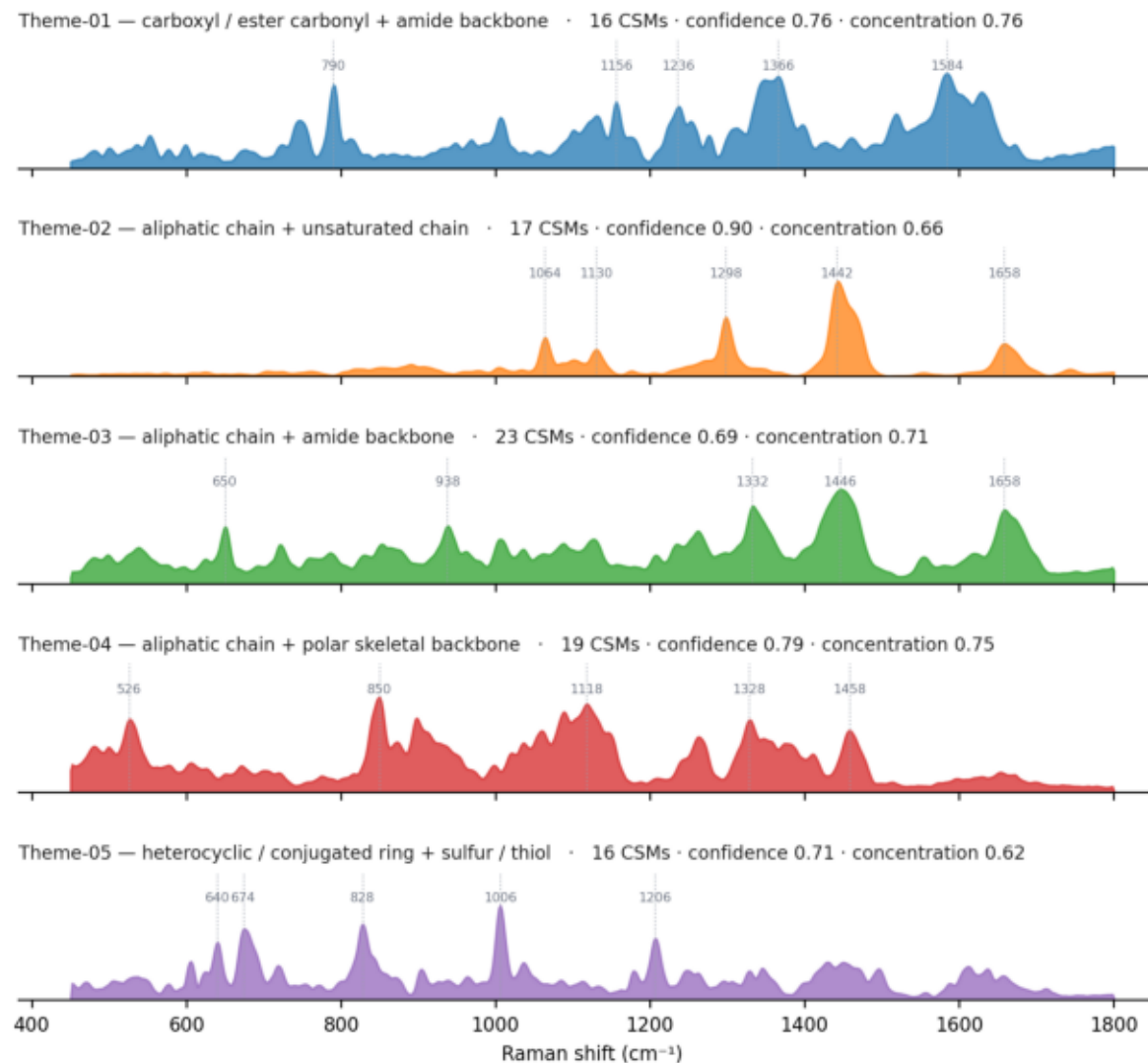
Model and K selection on label-free criteria — no chemistry label is visible at this stage



Model and K selection on label-free criteria. Stability is high wherever the membership is degenerate, so it must be read alongside information retained and the effective number of themes actually used; degenerate points are marked. Selected: archetypal analysis at K = 5.

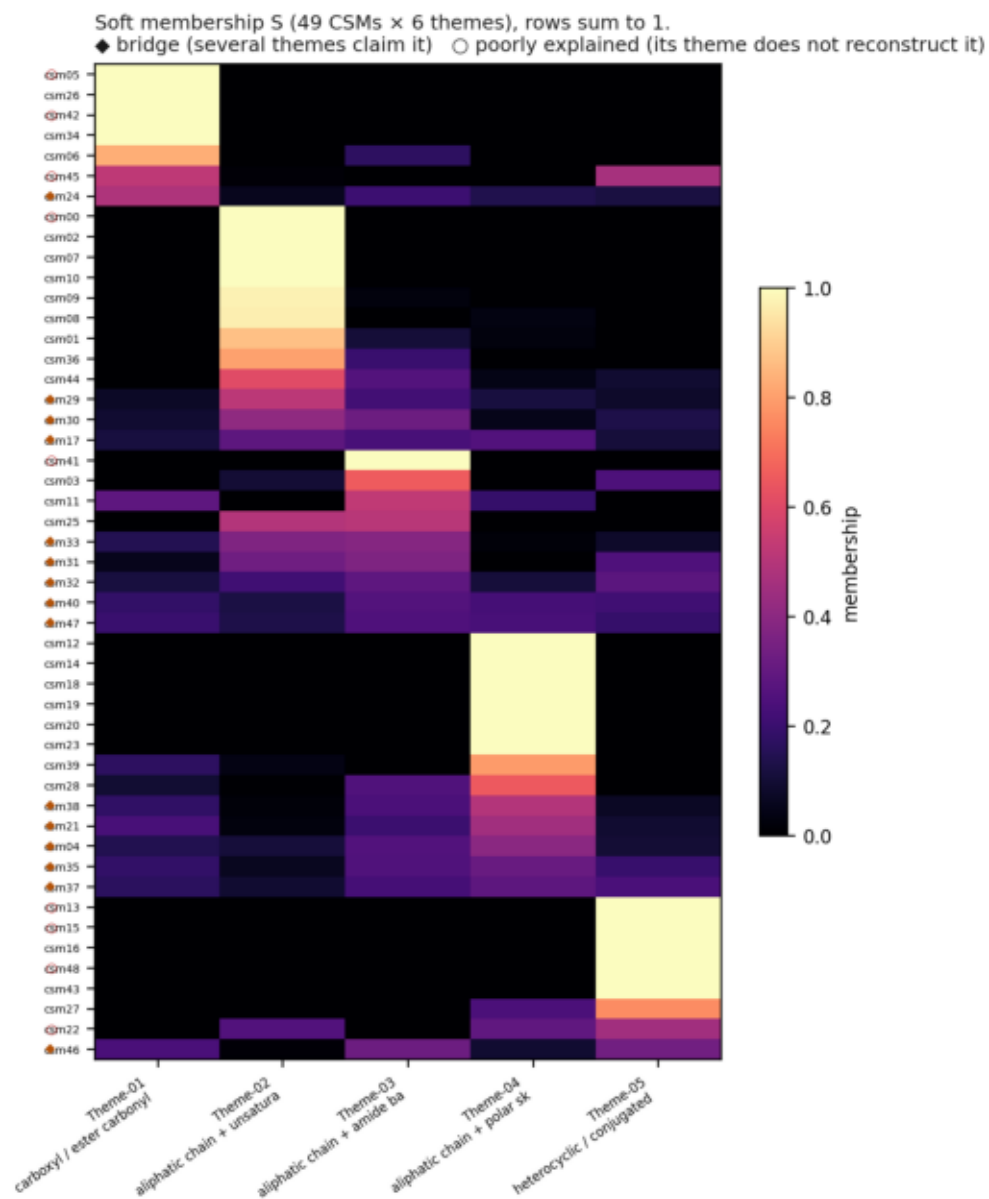
Figure 02

Theme spectra with their dominant bands. Names come from the bands, and only after validation.



The five theme spectra with their dominant bands. Names are derived from the bands and assigned only after validation. Theme-02 (1064, 1130, 1298, 1442, 1658 cm⁻¹) reproduces the acyl-chain-plus-cis-unsaturation signature that was the single accepted Phase 02 equivalence.

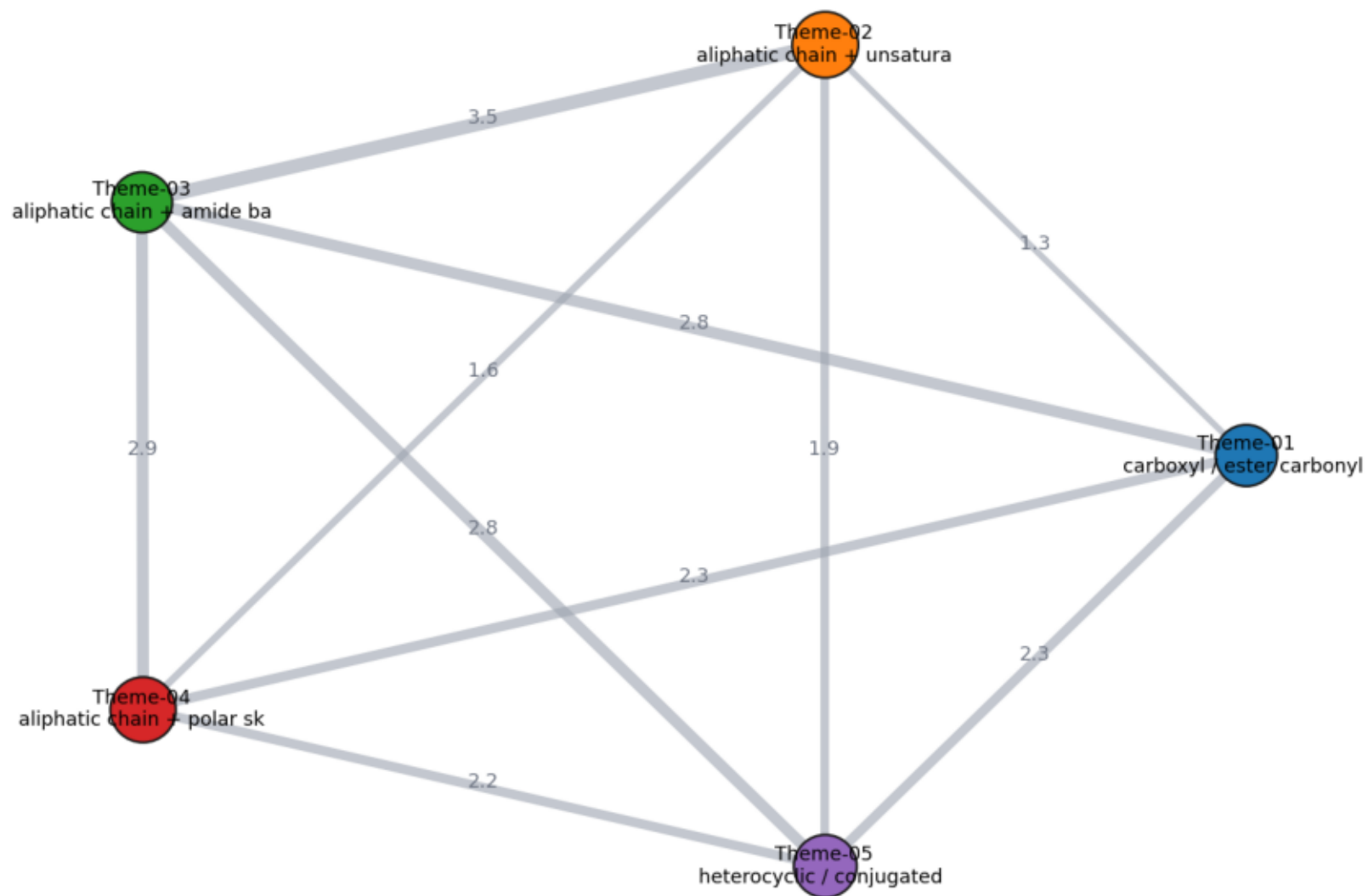
Figure 03



Soft membership S (49 CSMs × 5 themes), rows summing to one. Fifteen CSMs carry genuinely split membership and were left that way; nine are claimed by a theme that does not reconstruct them.

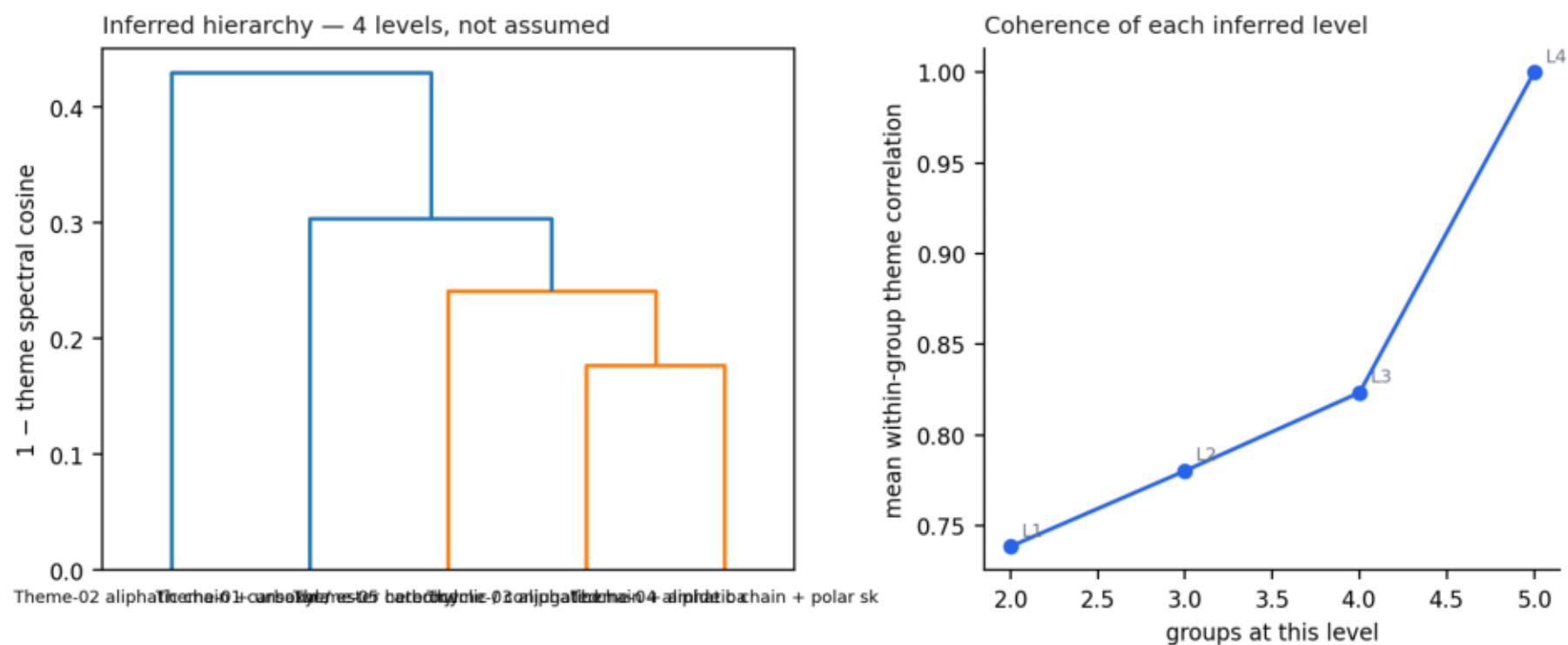
Figure 04

Theme overlap network. Edge weight is shared membership mass — themes are allowed to overlap, and the CSMs on those edges are the bridges.



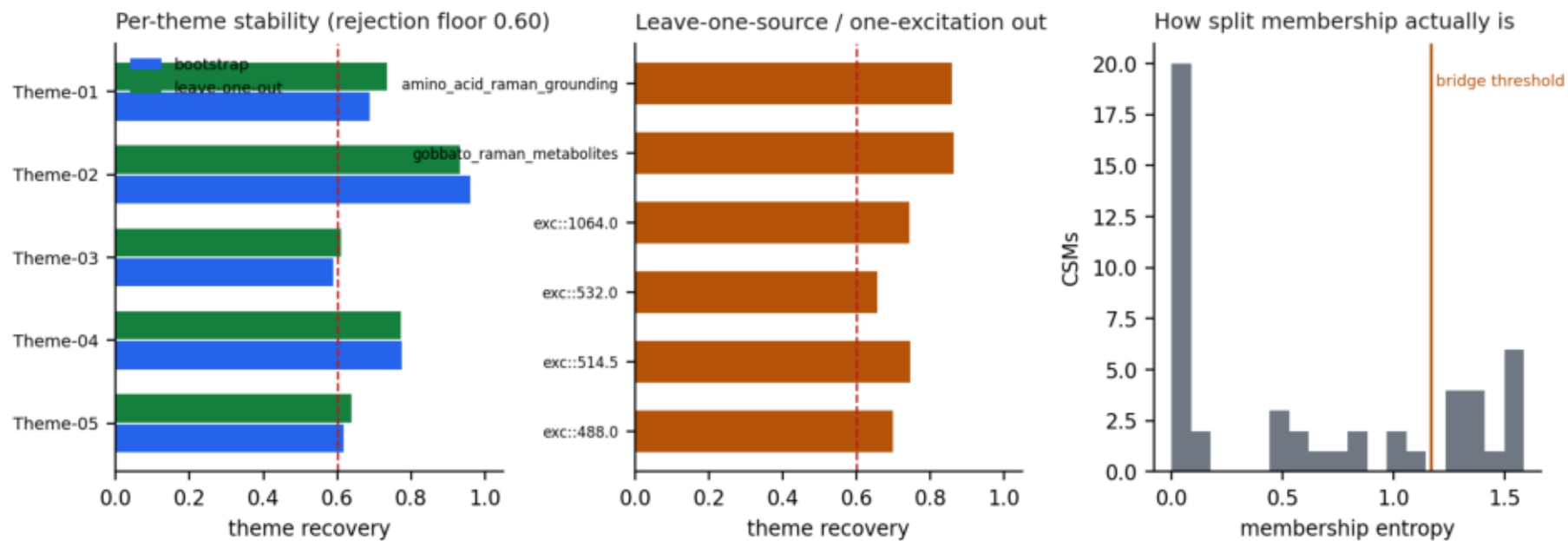
Theme overlap network — edge weight is shared membership mass. Themes are allowed to overlap, and the CSMs sitting on those edges are the bridges a hard assignment would misplace.

Figure 05



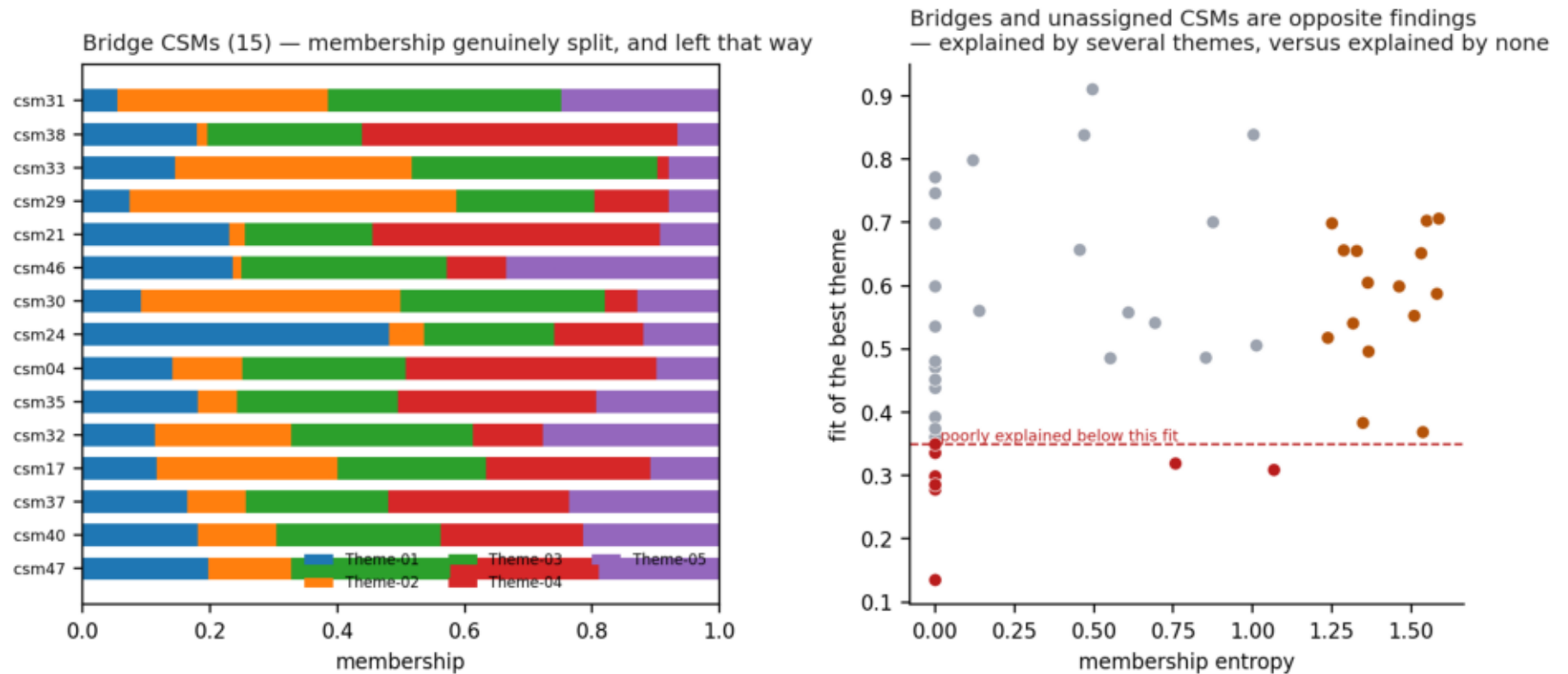
Inferred hierarchy. The number of levels was inferred rather than assumed. The top-level split separates aliphatic/unsaturated from polar/ring/carboxyl — the same hydrophobic/polar bipartition Phase 02.5 found independently in the CSM geometry, and the axis PCA found as its one reproducible component.

Figure 06



Per-theme stability against the 0.60 rejection floor, source and excitation holdouts, and the distribution of membership entropy. Theme-03 was rejected at 0.59 despite having the most supporting CSMs — membership breadth is not evidence.

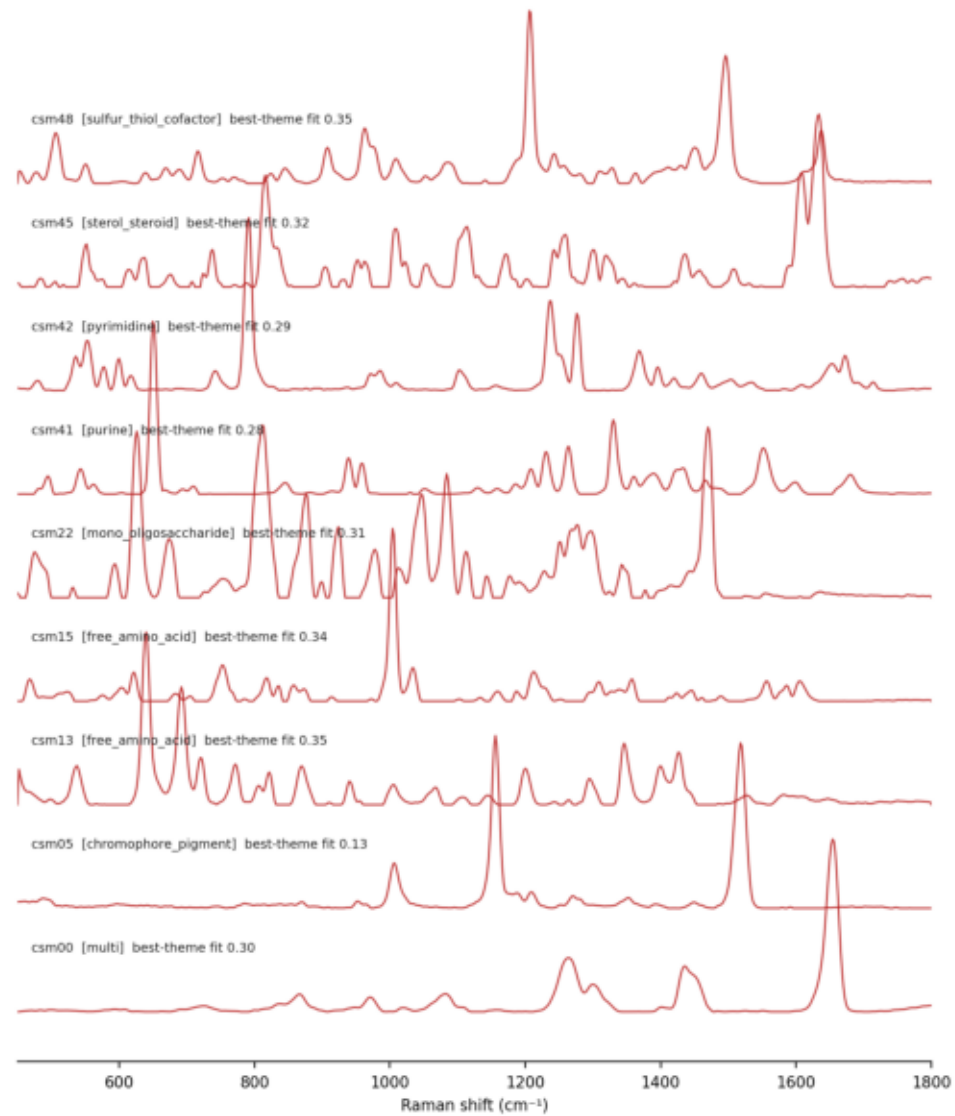
Figure 07



Bridge CSMs, and why bridges and poorly-explained CSMs are opposite findings: a bridge is well explained by several themes, a poorly-explained CSM by none.

Figure 08

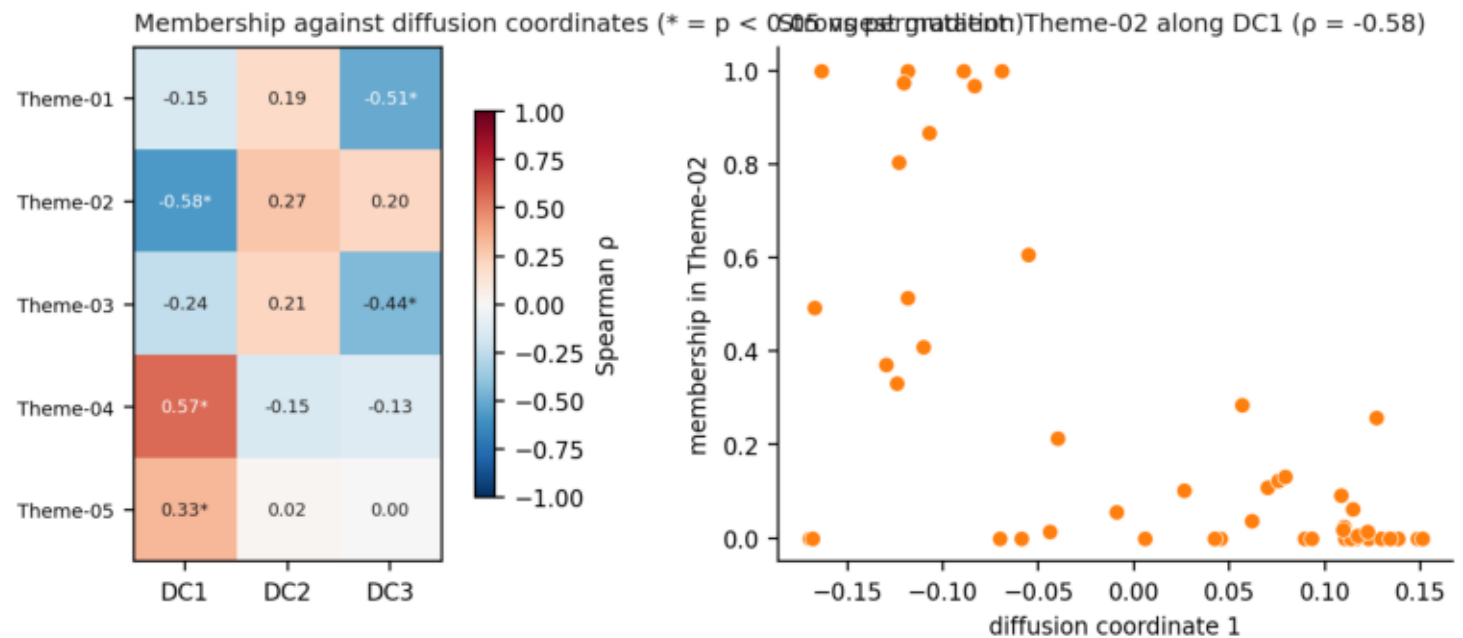
Poorly-explained CSMs — a theme claims them, but does not reconstruct them.
Recorded rather than absorbed: inventing a theme for an isolate is a motif borrowing foreign mass (L-03).



Poorly-explained CSMs. A theme claims each of them, but does not reconstruct them. Recorded rather than absorbed — inventing a theme for an isolate is a motif borrowing foreign mass (limitation L-03).

Figure 09

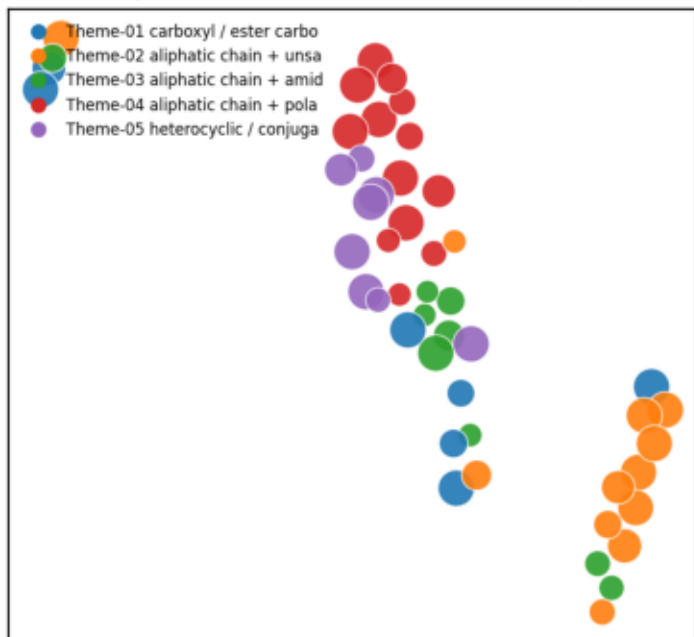
Phase 02.5 found a continuum; a theme layer that respects it shows membership varying smoothly rather than switching



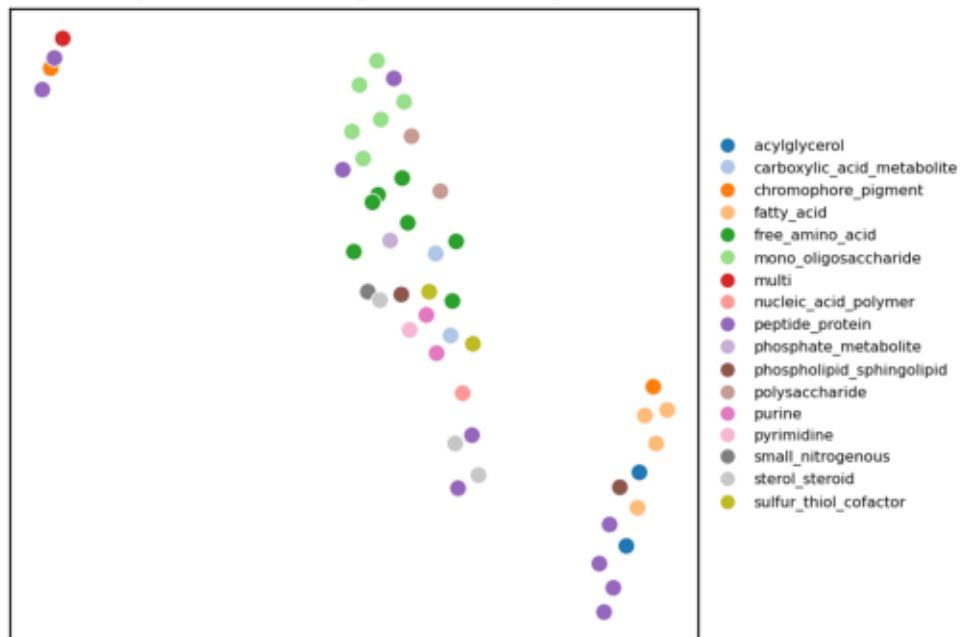
Theme membership against diffusion coordinates. Five of fifteen pairs are significant gradients against a permutation null: membership varies smoothly along the manifold rather than switching, which is what respecting the Phase 02.5 continuum means in practice.

Figure 10

Coloured by dominant theme; size = membership confidence

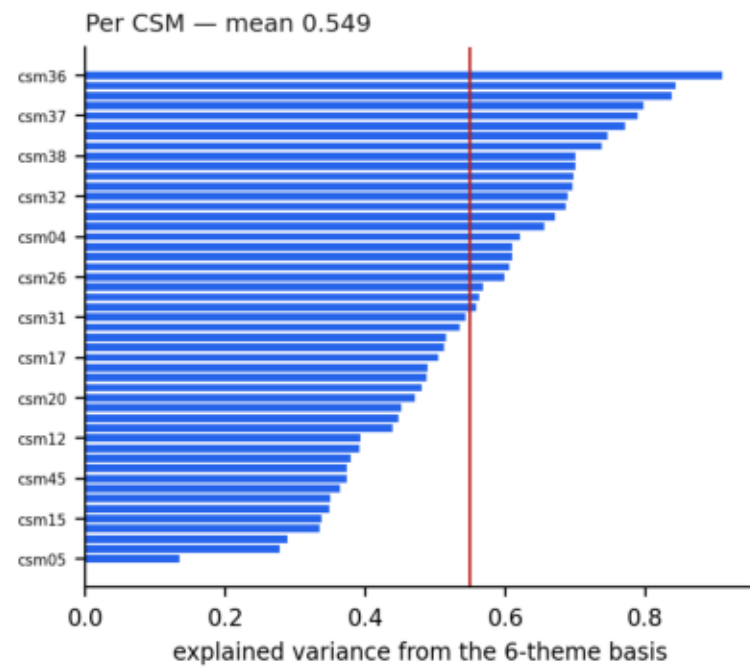


Coloured by curated chemistry — revealed only after the themes were fixed

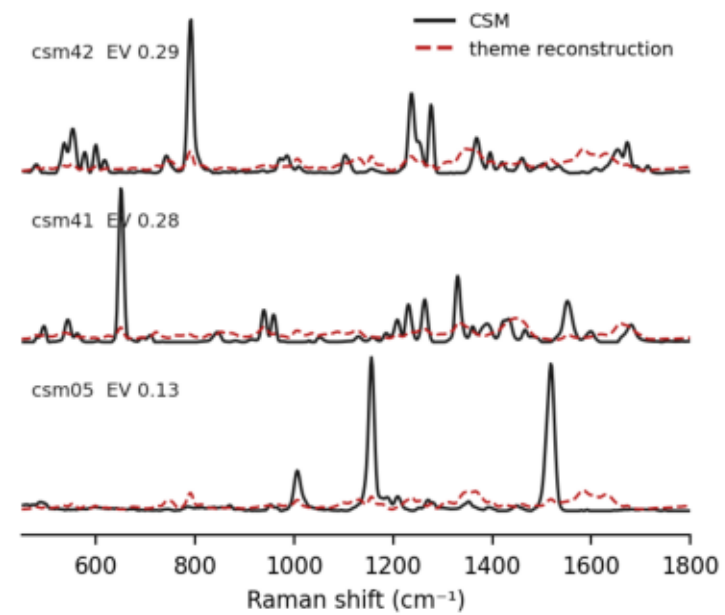


The CSM map coloured by dominant theme and, separately, by curated chemistry — the latter revealed only after the themes were fixed. Adjusted mutual information 0.157, $p < 0.0001$: better than chance, far from a re-derivation of the ontology.

Figure 11

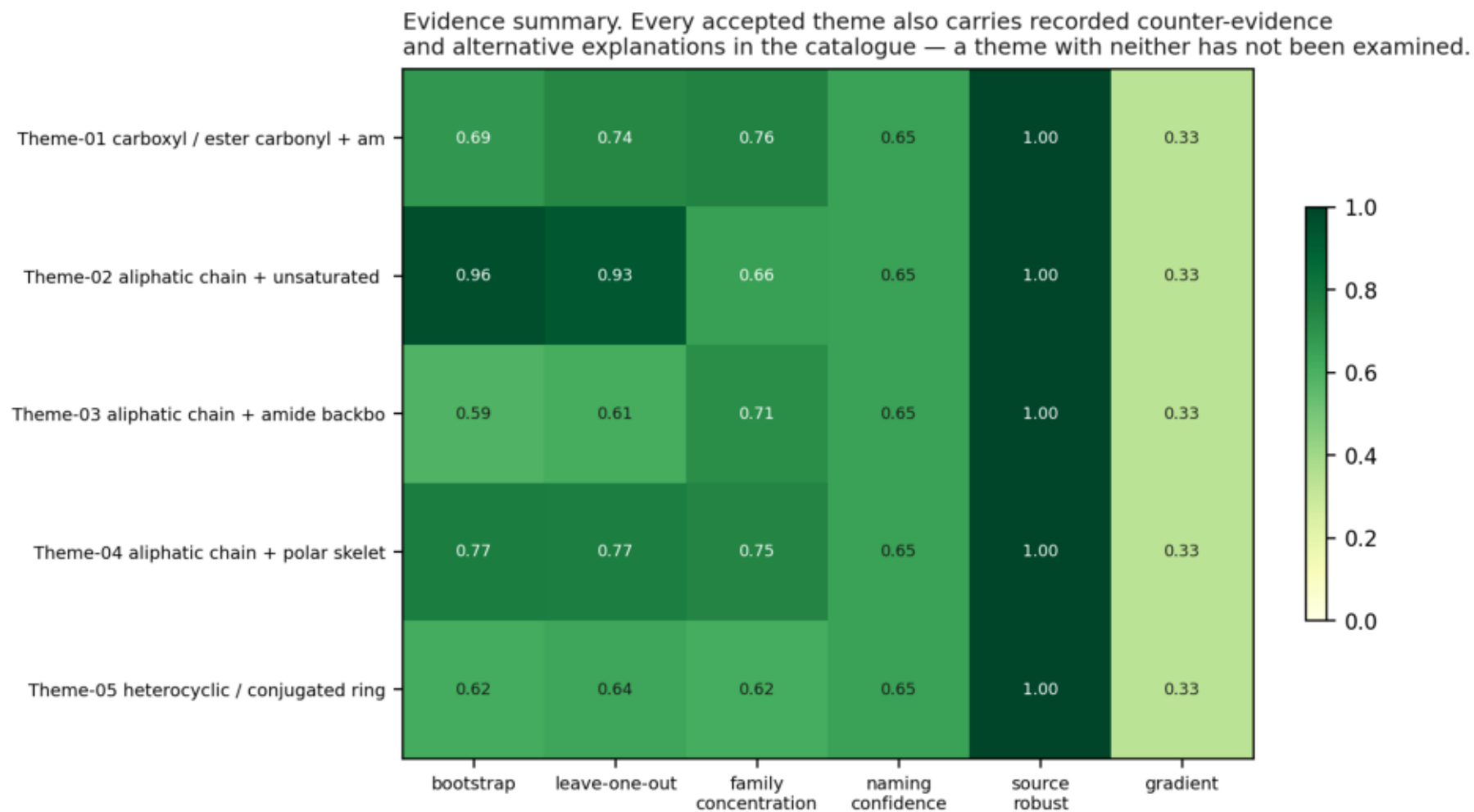


The three worst reconstructions — what the theme layer throws away



Reconstruction from the five-theme basis, per CSM, and the three worst cases. The theme basis explains 0.549 against the CSM basis's 1.000 at 9.8x compression — the price of abstraction, stated rather than smoothed over.

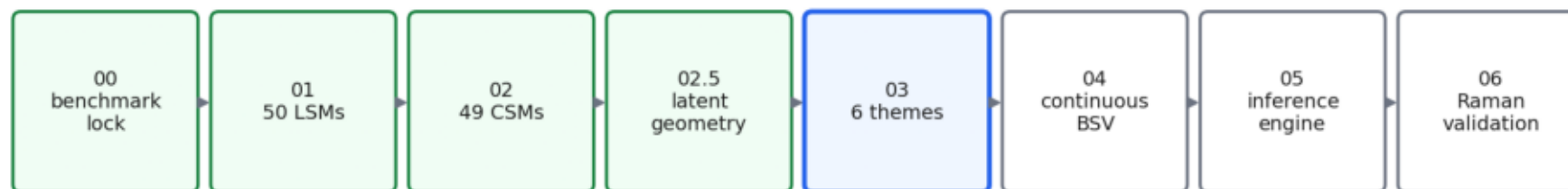
Figure 12



Evidence summary across the five themes. Every accepted theme also carries recorded counter-evidence and alternative explanations in the catalogue; a registry invariant enforces that a theme with neither has not been examined.

Figure 13

GAIRA V7 architecture after Phase 03



Phase 04 consumes: the membership matrix S (49×6), the 6-theme basis, the theme registry with its confidences and bridge annotations, and the unassigned list. BSV dimension = $K = 6$.

theme fingerprint f54d4835ffdf8aa2d50a4a203da0e8f4 · bootstrap 0.727 · AMI 0.157 · 15 bridges, 9 poorly explained